

De Broglie's Hypothesis of matter waves

4 times for 3 marks

From theory of radiation

$$E = h\nu = h \frac{c}{\lambda}$$

Einstein's mass energy relation

$$E = mc^2$$

$$h \frac{c}{\lambda} = mc^2$$

$$\lambda = \frac{h}{mc} = \frac{h}{p}$$

$$E = \frac{1}{2} mc^2 = \frac{(mc)^2}{2m} = \frac{p^2}{2m}$$

$$E = qV$$

$$p = \sqrt{2mE}$$

$$p = \sqrt{2mqV}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$$

Schrodinger's Time Dependant Equation

10 times for 5-8 marks

$$\Psi(x, t) = Ae^{i(kx - \omega t)}$$

Wave representation

Wave
properties
 ω & k

$$\begin{aligned}\omega &= 2\pi\nu = 2\pi \frac{E}{h} \rightarrow \omega = \frac{E}{\hbar} \\ k &= \frac{2\pi}{\lambda} = \frac{2\pi}{h} \cdot p \rightarrow k = \frac{p}{\hbar}\end{aligned}$$

Particle
properties
 E & p

Matter wave
representation

$$\Psi(x, t) = Ae^{\frac{i}{\hbar}(px - Et)}$$

w.r.t. 'x' twice

Differentiate

w.r.t. 't' once

$$\begin{aligned}\frac{\partial^2 \Psi(x, t)}{\partial x^2} &= \left(i \frac{p}{\hbar}\right)^2 Ae^{\frac{i}{\hbar}(px - Et)} = -\left(\frac{p}{\hbar}\right)^2 \Psi(x, t) \\ p^2 \Psi(x, t) &= -\hbar^2 \frac{\partial^2 \Psi(x, t)}{\partial x^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial \Psi(x, t)}{\partial t} &= -i \frac{E}{\hbar} Ae^{\frac{i}{\hbar}(px - Et)} = -i \frac{E}{\hbar} \Psi(x, t) \\ E \Psi(x, t) &= i\hbar \frac{\partial \Psi(x, t)}{\partial t}\end{aligned}$$

T.E=K.E + P.E

$$E = \frac{1}{2}mv^2 + V \rightarrow E = \frac{p^2}{2m} + V$$

$$\frac{1}{2m} p^2 \Psi + V \Psi = E \Psi$$

In Quantum mechanics
all observables are
operators hence
Operator equation
Needs to operates on
 Ψ

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V \Psi(x, t) = i\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

Schrodinger's Time Independent Equation

2 times for 5 marks

$$\Psi(x, t) = \Psi(x)\phi(t)$$

Doing variable separation

$$V = V(x)$$

Potential energy is function of position

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V \Psi(x, t) = i\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

$$i\hbar \Psi(x) \frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m} \phi(t) \frac{d^2 \Psi(x)}{dx^2} + V(x) \Psi(x) \phi(t)$$

Dividing by $\Psi(x)\phi(t)$

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\Psi(x)} \frac{d^2 \Psi(x)}{dx^2} + V(x)$$

position independent

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = E$$

Equating both sides to constant E

position dependent

$$E = -\frac{\hbar^2}{2m} \frac{1}{\Psi(x)} \frac{d^2 \Psi(x)}{dx^2} + V(x)$$

multiplying both sides with $\Psi(x)$

$$E \cdot \Psi(x) = -\frac{\hbar^2}{2m} \cdot \frac{d^2 \Psi(x)}{dx^2} + V(x) \Psi(x)$$

$$\frac{d^2 \Psi(x)}{dx^2} + \frac{2m}{\hbar^2} (E - V) \Psi(x) = 0$$

Energy of free particle in one dimensional potential well

5 times for 5 marks

$$\frac{d^2\psi(x)}{dx^2} - \frac{2mE}{\hbar^2}\psi(x) = 0$$

Time independent

Schrodinger's equation where we have substituted $V=0$

Assume $\frac{2mE}{\hbar^2} = k^2$

$$\frac{d^2\psi(x)}{dx^2} - k^2\psi(x) = 0$$

General solution for plane waves

$$\psi(x) = Ae^{ikx} + Be^{-ikx}$$

First boundary condition: $\psi(x) = 0$ at $x = 0$

$$0 = Ae^{ik \cdot 0} + Be^{-ik \cdot 0}$$

$$B = -A$$

Second boundary condition: $\psi(x) = 0$ at $x = L$

$$0 = Ae^{ikL} + Be^{-ikL}$$

$$A(e^{ikL} - e^{-ikL}) = 0$$

$$\sin kL = \frac{e^{ikL} - e^{-ikL}}{2i}$$

$$2i A \sin kL = 0$$

$$A \neq 0, \quad \sin kL = 0$$

$$kL = n\pi, \text{ Where } n=0,1,2,3,\dots$$

$$\sqrt{\frac{2mE}{\hbar^2}} L = kL$$

$$\sqrt{\frac{2mE}{\hbar^2}} L = n\pi$$

$$E = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

Squaring both sides, and after rearranging the terms